
AIM2 Course Project Proposal Title

Project Proposal (Week 4)

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Instructions (do not delete this box). This proposal is at most **2 pages of main text** (single-spaced, 12pt font, 1" margins). You may use a **3rd page for figures/tables** and have **unlimited references**. Keep the section headings below and replace the placeholder text with your own writing.

Hypothesis or Research Question

We investigate whether large language models can generate accurate, clinically coherent, and structured summaries of ICU patients by integrating fragmented clinical notes and structured EHR data from the MIMIC database. We further examine whether structured summaries improve information coverage and clarity compared to free-text summaries.

Background and Significance

Electronic health records (EHR) contain rich information about patients, but this information is often fragmented across multiple clinical notes, laboratory results, medication lists, and procedure records. As a result, clinicians must manually synthesize large volumes of heterogeneous data to understand a patient's clinical history and current status, which is time-consuming. Improving the way patient information is summarized and presented remains a critical challenge in healthcare, particularly in high-acuity settings such as intensive care units.

Prior studies have shown that domain-adapted large language models can achieve performance comparable to, or in some cases exceeding, that of medical experts in clinical text summarization tasks.[1] In addition, recent work has found no significant difference in quality between LLM-generated and physician-generated hospital discharge summaries, suggesting that LLMs may be capable of producing clinically meaningful patient summaries.[2]

However, most prior work focuses on single-document summarization or produces unstructured free-text outputs, leaving open questions about how well large language models can integrate fragmented, multi-modal EHR data. In this project, we aim to address these gaps using the MIMIC-III database to evaluate whether LLMs can synthesize unstructured clinical notes and structured EHR data into accurate, clinically coherent patient summaries and whether structured summaries provide advantages over free-text summaries in clarity and completeness.

Dataset: MIMIC-III

We use the Medical Information Mart for Intensive Care III (MIMIC-III) database, a large, publicly available, de-identified ICU dataset maintained by the MIT Laboratory for Computational Physiology. MIMIC-III contains detailed clinical data for more than 60,000 admissions to the ICU at Beth Israel Deaconess Medical Center between 2001 and 2012.

For this study, we utilize demographic information (age, sex, ethnicity), ICU stay characteristics, vital signs and bedside measurements, laboratory results, medication and infusion records, and

free-text clinical documentation such as nursing notes, physician progress notes, radiology reports, and discharge summaries.

MIMIC-III is appropriate for this project for several reasons. First, ICU patients generate high volumes of heterogeneous and time-sensitive clinical data, reflecting the real-world challenge of synthesizing fragmented EHR information. Second, the various structured data (labs, vitals, medications) and unstructured notes provide us with multi-modal data input and allow us to evaluate whether LLMs can integrate multi-modal inputs into coherent summaries. Finally, the dataset includes discharge summaries that can serve as a reference for evaluating information coverage and clinical coherence. Finally, its public availability and widespread use in clinical NLP research enable reproducibility and benchmarking.

Access to MIMIC-III requires completion of a Data Use Agreement and human subjects training. Preprocessing restricts the cohort to adult ICU stays (age ≥ 18) and accounts for age truncation in de-identified data. Tables are linked using `subject_id`, `hadm_id`, and `icustay_id`, and all structured events and notes are limited to the ICU admission window to prevent data leakage. Structured variables are cleaned through unit harmonization, missingness handling, and removal of duplicated values. Clinical notes are filtered and cleaned, with optional segmentation to meet model input limits. Finally, two parallel input formats are constructed: a concatenated free-text representation and a structured template format for controlled comparison.

Baseline Model

Our baseline is a zero-shot prompting approach, in which clinical notes from a patient’s ICU stay are concatenated and provided directly to a general-purpose open-source large language model (e.g., Llama 3 or Mistral) with a simple instruction to generate a patient summary. No retrieval, structured formatting, or in-context examples are provided. This setup reflects the most common starting point in clinical summarization research and isolates what a general-purpose LLM can produce without task-specific adaptation.[1] We evaluate the baseline output as a free-text summary compared against the corresponding MIMIC-III discharge summary using automated metrics.

Proposed Methodology

We propose a retrieval-augmented generation (RAG) pipeline with a clinically grounded structured output template. For each ICU admission, we first retrieve relevant clinical note segments and structured EHR data (laboratory results, vital signs, medications, and diagnoses) using a vector-based retrieval step (e.g., FAISS with a biomedical embedding model). Retrieved content is then passed to the LLM along with a structured prompt template modeled after the standard physician case presentation format: a clinical narrative (one-liner, history of present illness, hospital course), patient background (past medical history, medications, allergies, social history), current clinical status (active problems with management, pending workup), and key labs and imaging. This approach differs from the baseline in two key ways: (1) retrieval selects the most relevant information across both unstructured notes and structured tables rather than relying on naive concatenation, and (2) the structured template constrains the LLM output into an interpretable, clinically familiar format. Prior work has shown that RAG can improve EHR summarization accuracy and reduce hallucinations compared to direct prompting.[3]

For evaluation, we use a tiered approach. On the full cohort, we compute automated metrics including ROUGE-L (lexical overlap) and BERTScore (semantic similarity) against the discharge summary as a reference.[1] On a smaller subset, we assess clinical quality using the Provider Documentation Summarization Quality Instrument (PDSQI-9), a validated rubric that scores summaries on dimensions such as accuracy, organization, completeness, and clinical utility.[4] We supplement physician review with an LLM-as-a-Judge approach, which has demonstrated strong inter-rater reliability with human evaluators for this instrument.[5] The primary comparison is between the baseline free-text summary and the proposed structured summary across both automated and clinical quality metrics.

Resources

We will use the de-identified MIMIC dataset via PhysioNet access, focusing on ICU clinical notes and a limited set of EHR tables (labs, vitals, medications, procedures, diagnoses, and outcomes). Data extraction and cohort construction will be done with SQL with preprocessing in Python using pandas/numpy. All work will follow course and institutional policies for handling de-identified clinical data.

Our modeling stack will use PyTorch and Hugging Face (Transformers/Datasets/Accelerate), with optional parameter-efficient fine-tuning (LoRA/PEFT) if needed. We will implement simple baselines and metrics with scikit-learn, and optionally use a retrieval component (e.g. FAISS) to select relevant snippets/events for each summary field before generation. Development will be in Jupyter with GitHub for version control. We will prototype in Google Colab and run larger experiments on available GPUs; The primary pipeline will be reproducible with open-source models locally.

Challenges and Contingency Plans

Clinical notes are long and noisy, which can exceed context limits and reduce summary quality. We will control scope by restricting to a defined time window (e.g., first 24–48 hours) and high-yield note types, and we will use retrieval or chunk-and-aggregate summarization if inputs remain too large.

Hallucinations are a key risk for clinical summarization. We will require evidence links for generated fields (note/timestamp/snippet or structured event references) and add consistency checks against structured tables where possible. If hallucinations remain high, we will constrain outputs to a fixed schema and shift toward an extract-then-generate approach.

Compute and engineering constraints may limit fine-tuning and full EHR integration. We will start with prompting + retrieval and only use LoRA/PEFT if it provides clear gains; if resources are tight, we will run on a smaller stratified cohort and a reduced template. If structured-table integration becomes too time-consuming, we will deliver a notes-only system as the core baseline and treat structured integration as an extension.

Timeline and Milestones

Week 4–5: Complete data preprocessing pipeline, and construct ICU-level structured and unstructured data bundles

Week 6–7: Develop baseline LLM summarization pipeline

Week 8–9: Conduct baseline experiments and midterm presentation

Week 10–11: Perform full-scale experiments, comparative analysis, and robustness testing.

Week 12: Finalize experiments, complete analysis, and prepare final report and presentation

References

- [1] Dave Van Veen and Cara Van Uden. Adapted large language models can outperform medical experts in clinical text summarization. *Nature medicine*, 30(4):1134–1142, 2024.
- [2] Christopher Y. K. Williams and Charumathi Raghu Subramanian. Physician- and large language model-generated hospital discharge summaries. *JAMA Intern Med*, 185(7):818–825, 2025.
- [3] Mohammad Alkhalaf, Ping Yu, Mengyang Yin, and Chao Deng. Applying generative ai with retrieval augmented generation to summarize and extract key clinical information from electronic health records. *Journal of Biomedical Informatics*, 156:104662, 2024.
- [4] Emma Croxford, Yanjun Gao, Nicholas Pellegrino, Karen Wong, Graham Wills, Elliot First, Miranda Schnier, Kyle Burton, Cris Ebby, Jillian Gorski, et al. Development and validation of the

provider documentation summarization quality instrument for large language models. *Journal of the American Medical Informatics Association*, 32(6):1050–1060, 2025.

- [5] Emma Croxford, Yanjun Gao, Elliot First, Nicholas Pellegrino, Miranda Schnier, John Caskey, Madeline Oguss, Graham Wills, et al. Evaluating clinical ai summaries with large language models as judges. *NPJ Digital Medicine*, 8(1):640, 2025.